

Hamiltonian Edge Separation in Barnette Graphs: Certified Extremal Values and a Double-Ladder Family

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Draft of August 1, 2026

Abstract

For a Hamiltonian graph G , we study the minimum number $\text{hsep}(G)$ of Hamiltonian cycles needed to distinguish every ordered pair of distinct edges: for each (e, f) , some selected cycle must contain e and avoid f . We report an exhaustively certified finite computation over all 22,263 non-isomorphic Barnette graphs of orders at most 36. The resulting extremal values are independently checked by explicit Hamiltonian-cycle covers and matching lower certificates or complete-universe exclusion. In particular, the maximum at order 36 is 49 and is attained uniquely by the graph with canonical hash `2f96ada16c46cd2bd038b97d5af44f46ed14522cba1edf3fa041d66e02107bcc`. The unique maximizers at orders divisible by four form, over the certified range, a double-ladder family linked by a fixed four-vertex square insertion. We give an abstract definition of this family, prove that every member with odd positive parameters is a Barnette graph, and record certified structural and Hamiltonian-cycle patterns. A closed formula suggested by the certified hsep values is stated only as a conjecture. In a prediction-locked test, three subsequently constructed members $(D(9,9))$, $(D(11,9))$, and $(D(11,11))$ match both the Hamiltonian-cycle and hsep formulas exactly; these graph-specific certificates make no extremal claim at their orders. The terminology and its relationship to prior separation notions remain subject to a systematic literature review.

1 Introduction

A family of Hamiltonian cycles can be viewed as a binary measurement system on the edges of a graph: an edge receives one bit for every cycle, indicating whether the cycle contains it. Requiring every ordered pair (e, f) to be separated asks for a coordinate at which the signature of e is one and that of f is zero. This requirement is stronger than mere distinctness of the signatures and leads naturally to an antichain condition.

This manuscript develops that question for Barnette graphs, namely simple, planar, bipartite, cubic, and 3-connected graphs. Its first purpose is to state precisely what has been established by a finite computer-assisted proof. Its second purpose is to isolate a recurrent double-ladder structure among the certified extremal examples. The manuscript deliberately distinguishes four levels of evidence:

- (i) ordinary mathematical propositions proved in the text;
- (ii) finite computational results supported by explicit, independently checkable certificates;
- (iii) structural observations verified on the certified finite data; and
- (iv) empirical formulas and conjectures beyond the certified range.

The certified computation gives all extremal values from order 8 through order 36 whenever the Barnette census is nonempty. The order-10 census is empty and odd orders are impossible for cubic graphs. At the nonempty orders, exact upper certificates are Hamiltonian edge-separating families, while lower bounds are supplied either by packing requirements or by exhaustive exclusion over a

certified complete Hamiltonian-cycle universe. Importantly, no greedy cover size is used as an exact value.

The structurally recurring examples are denoted by $D(a, b)$. Two ladder strips are joined by four fixed terminal edges. The certified unique maximizers at orders (8,12,16,20,24,28,32,36) are respectively

$$D(1, 1), D(3, 1), D(3, 3), D(5, 3), D(5, 5), D(7, 5), D(7, 7), D(9, 7).$$

The tied extremal graphs at orders (22,26,30,34) have a different face structure and are not forced into this uniform family.

Literature-review status. TODO: A systematic peer-reviewed literature review is required before any novelty claim is made. It must cover separating and completely separating systems, edge separation by cycles, Hamiltonian-cycle covers, Barnette graph generation and census results, and reducible configurations or expansion operations. The notation hsep and the descriptive name used here are therefore provisional.

The remainder of the paper defines the invariant and its certificate logic, describes the census and verification pipeline, states the certified extremal sequence, analyzes the double-ladder family, and separates established results from open conjectures.

2 Definitions and certificate principles

All graphs in this manuscript are finite and undirected. A Hamiltonian cycle is regarded as an undirected edge set; reversal and cyclic rotation do not produce new cycles.

Definition 2.1 (Barnette graph). A *Barnette graph* is a simple, planar, bipartite, cubic, and 3-vertex-connected graph.

Let $\mathcal{H}(G)$ denote the set of undirected Hamiltonian cycles of a Hamiltonian graph G .

Definition 2.2 (Hamiltonian edge separation). For distinct edges $e, f \in E(G)$, a Hamiltonian cycle C *covers* the ordered requirement (e, f) if $e \in C$ and $f \notin C$. A family $\mathcal{C} \subseteq \mathcal{H}(G)$ is *Hamiltonian edge-separating* if it covers every ordered pair of distinct edges. Define

$$\text{hsep}(G) = \min \left\{ |\mathcal{C}| : \begin{array}{l} \mathcal{C} \subseteq \mathcal{H}(G), \\ \text{for all distinct } e, f \in E(G), \text{ some } C \in \mathcal{C} \text{ has } e \in C, f \notin C \end{array} \right\}. \quad (1)$$

Definition 2.3 (Barnette extremal sequence). For an order at which the relevant values are defined, put

$$M_B(n) = \max\{\text{hsep}(G) : G \text{ is a Barnette graph and } |V(G)| = n\}. \quad (2)$$

If no Barnette graph of order n exists, $M_B(n)$ is undefined. In the certified range every graph in every nonempty census is Hamiltonian, so equation (2) is unambiguous there.

Given an ordered family $C_1, \dots, C_k$, associate with an edge e its incidence signature

$$\sigma(e) = \{i : e \in C_i\} \subseteq \{1, \dots, k\}.$$

Proposition 2.4 (Signature antichain). A family $C_1, \dots, C_k$ is *Hamiltonian edge-separating* if and only if the edge signatures form an *antichain*: for all distinct e, f , neither $\sigma(e) \subseteq \sigma(f)$ nor $\sigma(f) \subseteq \sigma(e)$.

Proof. The requirement (e, f) is covered exactly when some index belongs to $\sigma(e) \setminus \sigma(f)$. Covering both ordered requirements (e, f) and (f, e) is therefore equivalent to incomparability of the two signatures. $\square$

This gives the usual Sperner-type counting obstruction. The computation also records a stronger incidence/LYM bound that uses cubicity, $|E(G)| = 3|V(G)|/2$, and the fact that every Hamiltonian cycle has exactly $|V(G)|$ edges. These general bounds are kept separate from graph-specific optimality certificates.

Lemma 2.5 (Primal and packing certificates). *If k explicit Hamiltonian cycles cover every ordered edge-pair requirement, then $\text{hsep}(G) \leq k$. If there are k requirements such that every Hamiltonian cycle covers at most one of them, then $\text{hsep}(G) \geq k$. Consequently, matching certificates prove $\text{hsep}(G) = k$.*

Proof. The first statement is the definition. For the second, each of the k requirements must be assigned to a selected cycle that covers it, while no selected cycle can receive two requirements. Hence at least k cycles are necessary. $\square$

When no matching packing is available, a lower bound in this project is called independently certified only if a standalone verifier checks a complete Hamiltonian-cycle universe and exhaustively excludes every cover below the claimed value.

3 Computational method

3.1 Census generation and graph validation

Complete non-isomorphic censuses were generated with plantri 5.8 (Brinkmann et al., n.d.). For target order n , the dual generation command has the form

$$\text{plantri -b -c3 -d T -}, \quad T = n/2 + 2.$$

The binary `planar_code` stream retains the rotation system. Each record was checked for the requested order and edge count, simplicity, cubicity, bipartiteness, planarity, 3-connectivity, and Euler and face arithmetic. Canonical hashes were computed from the sphere embedding; graph6, planar code, canonical edge lists, rotations, and faces were retained in the certificates. Generation indices were required to be contiguous and canonical hashes unique. Census completeness was never inferred from a retained candidate subset.

The plantri command, version, executable digest, environment, worker count, and runtime were recorded per order. At most four workers were used. Expensive stages used deterministic ordering, atomic checkpoints, and resumable output.

3.2 Hamiltonian cycles and separating covers

For a cubic graph of order n , there are $m = 3n/2$ edges and therefore $m(m-1)$ ordered edge-pair requirements. At order 36 this is $54 \cdot 53 = 2,862$ requirements per graph. Every emitted cycle was checked for closure, vertex uniqueness, spanning, and edge validity before it was used as a set-cover column.

For the smaller censuses, the complete undirected Hamiltonian-cycle universe was enumerated for every graph and the set-cover problem was solved exactly. For larger censuses, the pipeline first constructed deterministic separating covers for every graph and ranked the difficult tail. Once a

candidate value k had an exact lower and upper certificate, a competitor needed only a verified cover of size at most k , or at most $k - 1$ to establish uniqueness. Complete-universe enumeration and exact optimization were then restricted to graphs that could still attain the maximum. A timeout or incomplete universe was not accepted as a conclusion.

3.3 Three levels of evidence

The proof pipeline deliberately separates the following objects.

Solver output. A proposed cycle, cover, optimum, or infeasibility result produced by a search or optimization implementation.

Emitted certificate. Canonically encoded graphs and explicit cycles, together with either packing requirements or a certified complete cycle universe suitable for exhaustive exclusion.

Independent verification. A standalone Python standard-library program reparses the artifacts, checks graph and cycle structure, evaluates every ordered requirement, checks lower certificates, order summaries, and SHA-256 manifests, and does not import the project, NetworkX, plantri, SAT, or optimization packages.

For every nonmaximizer the final package contains an independently verified cover strictly smaller than the maximum at that order. For every maximizer it contains an exact primal certificate and either a matching packing or a complete-universe exhaustive lower proof. Thus a greedy witness count is never reported as $\text{hsep}(G)$.

4 The certified finite extremal sequence

Computational Result 4.1 (Certified values through order 36). The complete Plantri 5.8 censuses and independent certificate verification give the values in table 1. Across the nonempty orders, 22,263 non-isomorphic graphs were audited. Every maximum and tie count was independently verified, and every nonmaximizer has a verified separating cover strictly below the maximum of its order.

Every cubic graph has even order, since the degree-sum identity gives $3|V| = 2|E|$. Hence all odd orders from 9 through 35 are impossible. The complete order-10 Plantri census is empty. These undefined entries are not optimization failures.

The order-36 statement is especially useful as a regression target:

$$M_B(36) = 49,$$

with a unique maximizer of canonical hash

2f96ada16c46cd2bd038b97d5af44f46ed14522cba1edf3fa041d66e02107bcc.

Its complete universe contains 68 undirected Hamiltonian cycles. A 49-cycle primal certificate and a 49-requirement packing certificate prove its exact value. All 15,373 competitors have independently checked covers of size at most 48.

Matching packings certify the displayed maximizers at orders 8, 12, 16, 20, 24, 28, 32, 36. At orders 14, 18, 22, 26, 30, 34, at least one displayed exact result uses exhaustive set-cover exclusion over a certified complete cycle universe instead. This distinction concerns the form of the lower certificate, not the exactness of the result.

Status note. Theorem 4.1 is an exhaustive finite computational theorem about the listed orders. It is not a proof of any formula for arbitrary n , nor does it assert anything about orders 38 or 40.

Table 1: Certified Barnette Hamiltonian edge-separation extrema. The last column is the number of undirected Hamiltonian cycles in each maximizer; within every tied order all maximizers have the displayed count.

n	graphs	$M_B(n)$	maximizers	Hamiltonian cycles
8	1	6	1	6
12	1	8	1	8
14	1	10	1	12
16	2	12	1	14
18	2	10	1	16
20	8	17	1	20
22	8	13	2	28
24	32	24	1	30
26	57	18	5	40
28	185	31	1	40
30	466	25	3	60
32	1,543	40	1	54
34	4,583	32	7	80
36	15,374	49	1	68

5 The double-ladder family

Let $L_a = P_{a+1} \square K_2$ be a ladder with vertices A_i^r , where $0 \leq i \leq a$ and $r \in \{0, 1\}$. Its rungs are $A_i^0 A_i^1$, and its rails are $A_i^r A_{i+1}^r$. Define a second, disjoint ladder L_b on vertices B_j^r .

Definition 5.1 (Double ladder). For positive integers a, b , the graph $D(a, b)$ consists of L_a and L_b together with the four connector edges

$$A_0^0 B_0^0, \quad A_0^1 B_b^0, \quad A_a^0 B_0^1, \quad A_a^1 B_b^1. \quad (3)$$

Thus

$$|V(D(a, b))| = 2(a + b) + 4.$$

5.1 The square insertion

Suppose $uvyxu$ is a facial quadrilateral, so that uv and xy are opposite edges. Replace uv by $upqv$, replace xy by $xrsy$, and add pr and qs . The four new vertices induce the facial cycle $pqsrp$. In a ladder cell this operation replaces one cell by three and increases the ladder length by two.

Lemma 5.2 (Preservation under facial square insertion). *Let G be a simple, planar, bipartite, cubic, 3-connected graph, and apply the preceding operation to opposite edges of a facial quadrilateral. The resulting graph G^+ has the same five properties.*

Proof. The four terminals are distinct and the inserted vertices are new, so simplicity is immediate. Drawing the three-cell ladder inside the old face proves planarity. Each old terminal loses one edge and gains one edge; every new vertex has two neighbours on the inserted 4-cycle and one old terminal. Thus cubicity is preserved. If the facial quadrilateral is $uvyxu$, then u, y lie in one bipartition class and v, x in the other. The unique alternating coloring along the two new length-three paths makes both pr and qs bichromatic, proving bipartiteness.

It remains to prove 3-connectivity. Let $S \subseteq V(G^+)$ with $|S| \leq 2$, and call p, q, r, s the new vertices. If S contains no new vertex, paths in the connected graph $G - S$ lift through the two intact replacement paths; the intact inner 4-cycle is attached to every surviving terminal. If S contains exactly one new vertex, at most one old vertex is deleted. The graph obtained from G by deleting that old vertex is 2-connected and hence has no bridge. Delete, if necessary, the one old edge whose replacement path contains the removed new vertex. The remaining old vertices are still connected, and the other three new vertices form a path attached to at least two surviving terminals. Finally, if two new vertices are deleted, no old vertex is deleted. A 3-vertex-connected graph is 3-edge-connected, so $G - \{uv, xy\}$ is connected. Every remaining new vertex has its own terminal edge into this connected old core. Hence $G^+ - S$ is connected in all cases, and G^+ is 3-connected. $\square$

Theorem 5.3. *For all odd integers $a, b \geq 1$, the graph $D(a, b)$ is a Barnette graph.*

Proof. The graph $D(1, 1)$ is isomorphic to the cube. Increasing either odd parameter by two is exactly the facial square insertion of theorem 5.2 in a cell of the corresponding ladder. Starting from $D(1, 1)$, repeated insertions therefore reach every pair of positive odd parameters while preserving all Barnette properties. $\square$

5.2 Certified occurrences and face structure

Computational Result 5.4 (Certified double-ladder recurrence). Exact graph isomorphisms identify the unique maximizers at orders divisible by four as follows:

n	8	12	16	20	24	28	32	36
G_n	$D(1, 1)$	$D(3, 1)$	$D(3, 3)$	$D(5, 3)$	$D(5, 5)$	$D(7, 5)$	$D(7, 7)$	$D(9, 7)$.

All seven consecutive transitions have reconstructible expansion certificates whose reconstructed edge set is exactly the certified target and whose reduced canonical hash is exactly the certified source.

Proposition 5.5 (Face pattern in the recursive embedding). *For odd positive a, b , the embedding obtained from the cube by the square insertions above has $a + b$ designated ladder-cell faces, all quadrilateral, together with two faces of length $a + 3$ and two faces of length $b + 3$. When $a = 1$ or $b = 1$, the corresponding two cap faces are themselves quadrilateral.*

Proof. For $D(1, 1)$, the two designated cells and four cap faces are the six faces of the cube. Increasing a by two replaces one designated cell by three, adding two quadrilateral cells. The two cap boundaries running along the modified (a)-ladder each acquire two edges; the other two cap boundaries are unchanged. This is precisely the claimed update. The argument for b is identical, so induction proves the formula. $\square$

For the certified members from $D(3, 3)$ onward, the two sets of designated cells are exactly the two path components of the quadrilateral face-adjacency graph. At orders 8 and 12, additional quadrilateral cap faces make this face-adjacency description degenerate, although the abstract double-ladder definition remains valid.

6 Hamiltonian-cycle structure

The complete Hamiltonian-cycle universes of the five primary double-ladder members from orders 20 through 36 were encoded using four connector bits, rung bits, two rail strings per ladder, four-bit

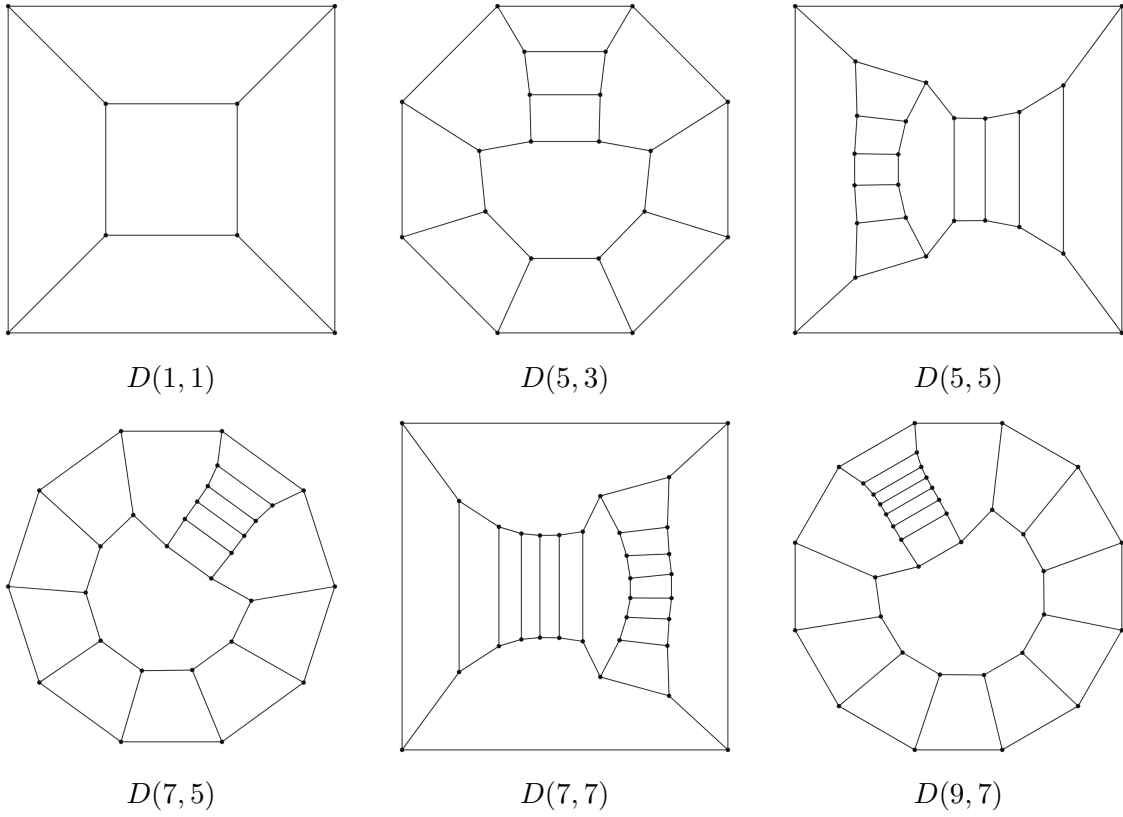


Figure 1: Selected certified double-ladder maximizers. Layouts were produced from the certified planar embeddings; geometry is used only for presentation, while all structural identifications were made from abstract edges and rotation systems.

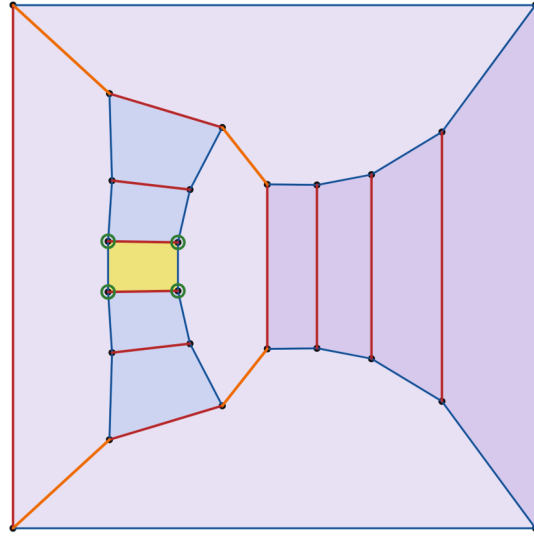


Figure 2: Annotated $D(5, 5)$. Blue edges are rails, red edges are rungs, orange edges are terminal attachments, and the highlighted four-vertex tile is a certified square insertion. The inverse operation deletes the tile and restores the two opposite rail edges.

cell states, and adjacent-cell transitions. Complete universes and cycle counts are also certified for the three smaller, face-degenerate members, but those three were not included in the detailed tile-state artifact. The resulting finite pattern is particularly simple.

Computational Result 6.1 (Certified cycle counts). For each certified graph in

$$D(1, 1), D(3, 1), D(3, 3), D(5, 3), D(5, 5), D(7, 5), D(7, 7), D(9, 7),$$

the complete Hamiltonian-cycle universe has size

$$|\mathcal{H}(D(a, b))| = ab + 5.$$

For each of the five state-encoded members from $D(5, 3)$ through $D(9, 7)$, every cycle belongs to exactly one of three observed classes: one all-rails cycle, ab cycles determined by one turn cell in each ladder, and four two-connector exceptions.

Here is the local state mechanism behind the finite classification. Number the connectors in the order displayed in equation (3). A cycle crosses the four-edge cut between the two ladders an even, nonzero number of times. With all four connectors present, the degree-two equations along one ladder force the top and bottom rail decisions to agree cell by cell. In every certified universe, excluding an internal closed component leaves either the all-rails state or a single omitted rail pair bounded by two selected rungs. The latter is a turn at one of the ladder cells. Two straight ladders give the all-rails cycle; two turned ladders give one cycle for each independent pair of turn positions.

With exactly two connectors, the four observed Hamiltonian connector words are

$$1100, \quad 0011, \quad 1010, \quad 0101,$$

each with one forced completion for odd ladder lengths. The remaining words 1001 and 0110 do not occur in the certified complete universes.

Observation 6.2 (Status of the general count). The emitted state data exhaustively establish theorem 6.1 for the eight analyzed members. They strongly suggest that the same degree recurrence is a symbolic proof for all positive odd a, b , but the present artifact package checks complete finite universes rather than a separately formalized parameterized recurrence. Accordingly, this manuscript does not yet promote $|\mathcal{H}(D(a, b))| = ab + 5$ to a theorem for the infinite family. The smaller certified members support the count but not the detailed state classification.

Proof TODO. Formalize the rail/rung degree recurrence for arbitrary odd lengths, prove that multiple turns create a separate cycle, and verify the six two-connector boundary cases without reference to enumerated examples.

7 Tied orders, empirical laws, and conjectures

7.1 Tied maximizers

The extremal behavior at orders congruent to two modulo four is structurally different from the uniform double-ladder recurrence.

Computational Result 7.1 (Certified tied-order structure). Every one of the 17 tied maximizers at orders (22,26,30,34) contains a 3-cycle component in the face-adjacency graph induced by quadrilateral faces, as well as at least one strict ladder component. The full quadrilateral component signatures are pairwise distinct within each order and reproduce the tie counts (2,5,3,7).

Twenty-five exact square-gadget certificates link these graphs to earlier examples. Thirteen links are literal ladder-cell insertions. In the other twelve, the same four-vertex gadget is inserted on two suitable disjoint edges of a 6-, 8-, or 10-face end-cap region. Thus these graphs are not asserted to be members of one $D(a, b)$ recurrence.

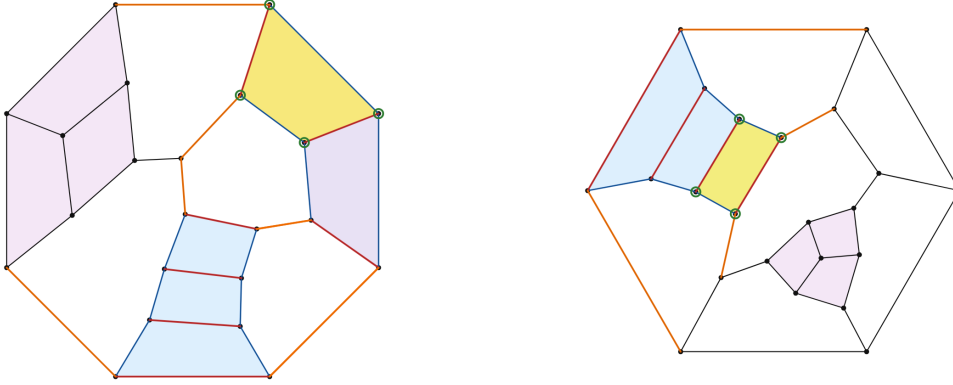


Figure 3: The two certified order-22 maximizers. Purple marks the quadrilateral 3-cycle component; blue and red distinguish ladder rails and rungs. Their different component signatures illustrate why the tied branch is not identified with a single double-ladder skeleton.

7.2 Empirical hsep formula

The six independently certified values from orders 16 through 36 agree with

$$\text{hsep}(D(a, b)) = \frac{ab + 2a + 2b + 3}{2}. \quad (4)$$

The supporting parameter pairs and values are

(a, b)	(3, 3)	(5, 3)	(5, 5)	(7, 5)	(7, 7)	(9, 7)
hsep	12	17	24	31	40	49.

The formula does not match the two degenerate initial members $D(1, 1)$ and $D(3, 1)$.

Conjecture 7.2 (Double-ladder hsep formula). *For odd $a, b \geq 3$, equation equation (4) holds.*

This conjecture is not a consequence of the finite expansion certificates. In the five primary packing certificates at orders 20 through 36, exactly $k - 2$ packing requirements have both edges internal to the ladders, while the remaining two have only their included edge internal. No invariant placement or certificate-lifting theorem has yet been found.

Conjecture 7.3 (Extremal continuation). *Balanced or nearly balanced double ladders continue to be extremal at larger orders divisible by four, and eventually are uniquely extremal there.*

Even if theorem 7.2 is true, it does not prove this extremal statement; all competing Barnette graphs at a given order would still need an upper certificate.

Computational Result 7.4 (Prospective double-ladder test). Before constructing or analyzing any new graph, the formulas predicted

G	$ \mathcal{H}(G) $	$\text{hsep}(G)$
$D(9,9)$	86	60
$D(11,9)$	104	71
$D(11,11)$	126	84.

The prediction lock was then left unchanged. Each graph was constructed from the certified $(D(9,7))$ member by successive four-vertex square insertions. For all three graphs, a path-based enumeration and an independent exhaustive perfect-matching enumeration produced the predicted complete Hamiltonian universe. Each predicted hsep value has an explicit primal cover and a matching packing of the same size, checked by a standard-library verifier.

Thus the two family formulas survive three genuinely prospective tests, but a finite continuation does not prove theorem 7.2. No order-40, order-44, or order-48 census or extremal certification was performed. In particular, this manuscript does not claim values for $M_B(40)$, $M_B(44)$, or $M_B(48)$. A separate linear fit to only the three tied-order values at $(26,30,34)$ predicts $M_B(38) = 39$; because the tied expansion graph branches and merges, that extrapolation is considered especially weak. No order-38 computation is reported here.

8 Reproducibility and independent verification

The source repository contains the producer, structural analysis, tests, and standalone-verifier source. Large immutable result packages remain outside the repository. The manuscript itself contains only the compact table and selected figure files needed to compile it.

The global standard-library verifier checked all 29 order packages from 8 through 36, including the 14 nonempty orders, 266 manifest-listed files, every cycle certificate, every ordered edge-pair cover, all lower certificates, and the global summaries. Its recorded runtime was 655.989 seconds. The producer runtimes recorded across the orders sum to 5,186.661 seconds; those values mix fresh production and the imported order-36 workflow and are provided as audit metadata rather than a hardware-independent benchmark.

The principal immutable identifiers used for this draft are

package	SHA-256 of SHA256SUMS.txt
certified sequence	2a0a27cbb2c5dcefc3173293dc1fbfb21b55a69cdd366da3ab3b30a2842affa
maximizer gallery	a7a3ec48ece3d06191e3543f996f23b5c2d978978dc9634dcff2679746c23aa4
ladder analysis	17b1a8219321fc70ac8146f17e29d8b836c90862ec96248429141869e252a67e
prospective double-ladder test	6bd843dfbe9f3eb1aa3213511ce28091efc7567f7d7565f0aa980943b05713a1

The ladder-analysis manifest contains 30 entries, all of which were rechecked before drafting. It binds the structural tool and module hashes to the analysis output without modifying the certified sequence or gallery inputs.

The repository test run associated with this draft collected 185 tests: 180 passed and five environment-gated external integrations were skipped. The skipped tests require explicitly configured plantri or cubhamg executables and do not replace the standalone certificate verification described above.

The prospective package locks its predictions in a read-only JSON record with SHA-256 **e8ba381dd163c59941db81f65b71b70d6d78cdb7cfd6f6a41ba4828fe25c89fb**. Its standalone verifier passed all three constructions, complete universes, structural classifications, primal covers, matching lower certificates, and local and global manifests.

The planar layouts in this manuscript derive from certified rotation systems using Gunnar Brinkmann’s external drawing program. The wrapper checked that the vertices and edges recovered from every generated drawing exactly match the corresponding certified graph. Use of that program does not imply endorsement, collaboration, or coauthorship.

Draft AI-assistance disclosure

AI tools assisted with software development, data analysis, drafting, and language editing. All mathematical certificate packages are independently checkable. The author remains responsible for the definitions, claims, proofs, source verification, interpretation of computational evidence, and final manuscript.

9 Conclusion

The certified computation determines $M_B(n)$ for every nonempty Barnette census from order 8 through order 36, including all ties and the unique order-36 value 49. Its proof objects are explicit: upper certificates are Hamiltonian-cycle families, and lower certificates are either packings or complete-universe exhaustive exclusions checked by a standalone verifier.

The finite maximizers reveal two distinct structural regimes. The unique maximizers at orders divisible by four lie in the explicitly defined double-ladder family and are connected by one facial square insertion. The tied orders instead exhibit a quadrilateral 3-cycle component and a branching collection of ladder and cap expansions. These facts are certified over the analyzed graphs. The proposed general Hamiltonian-cycle and hsep formulas, and the suggested extremal continuation, remain open until parameterized proofs and new order-level certificates are supplied. Three prediction-locked tests beyond the census range confirm the graph-specific formulas for $(D(9,9))$, $(D(11,9))$, and $(D(11,11))$, but do not establish that any of those graphs is extremal at its order.

The immediate mathematical priorities are to formalize the general ladder state recurrence, construct hsep certificates that lift under square insertion, and classify the tied boundary states. Before circulation as a research claim, the manuscript also requires expert review of the direct 3-connectivity proof and a systematic literature review of all related separation notions and graph operations.

References

Gunnar Brinkmann, Brendan McKay, and Heidi Van den Camp. *Guide to Using plantri (Version 5.8)*, n.d. URL <https://users.cecs.anu.edu.au/~bdm/plantri/>. Software guide distributed with plantri 5.8.